

Hanmeng (Harmony) Zhan

CONTACT INFORMATION	Computer Science Department 100 Institute Rd, Worcester, MA 01609, USA	hzhan@wpi.edu hanmengzhan.com
RESEARCH INTERESTS	Algebraic graph theory, discrete quantum walks, continuous quantum walks, graph spectra, association schemes, orthogonal polynomials, equiangular lines	
CURRENT POSITION	Assistant Professor Computer Science Department, Worcester Polytechnic Institute, Worcester, MA, USA	Aug 2023 - Present
PREVIOUS POSITIONS	Postdoctoral Fellow - Department of Mathematics, Simon Fraser University, Vancouver, BC, Canada Topic of projects: Spectral graph theory Supervisor: Bojan Mohar York Science Fellow - Department of Mathematics and Statistics, York University, Toronto, ON, Canada Topic of projects: Discrete quantum walks Supervisor: Ada Chan Postdoctoral Fellow - Centre de Recherches Mathématiques, Université de Montréal, Montréal, QC, Canada Topic of projects: Continuous quantum walks Supervisor: Luc Vinet	Jan 2022 - May 2023 Oct 2019 - Sep 2021 Oct 2018 - Sep 2019
EDUCATION	University of Waterloo , Waterloo, ON, Canada - Ph.D. Department of Combinatorics and Optimization, Faculty of Mathematics Thesis: <i>Discrete Quantum Walks on Graphs and Digraphs</i> Supervisor: Chris Godsil Thesis awards: - University Finalist for the Governor General's Gold Medal - Inaugural Mathematics Doctoral Prize - Master of Mathematics Department of Combinatorics and Optimization, Faculty of Mathematics Thesis: <i>Uniform Mixing on Cayley Graphs over $\mathbb{Z}_3^d$</i> Supervisor: Chris Godsil Thesis award: - Outstanding Achievement in Graduate Studies	May 2014 - Sep 2018 Sep 2012 - Apr 2014

- Bachelor of Arts Jan 2010 - Aug 2012
Department of Economics, Faculty of Arts
Thesis: *Second-Price Auction with Resale*
Supervisor: Philip Curry
Xiamen University, Xiamen, Fujian, China
- Bachelor of Economics Sep 2008 - Jun 2014
Department of Statistics, Faculty of Economics
Thesis: *Multi-Player Multi-State Quantum Games*
Supervisor: Zhengming Qian

PUBLICATIONS **Book**

1. C. Godsil, H. Zhan, *Discrete Quantum Walks on Graphs and Digraphs*. Cambridge University Press (2023).

Peer-Reviewed Papers

2. Q. Chen, C. Godsil, M. Sobchuk, H. Zhan. *Hamiltonians of bipartite walks*. Electronic Journal of Combinatorics (2024) 31(4): P4.10.
3. A. Chan, H. Zhan, *Pretty good state transfer in discrete-time quantum walks*. Journal of Physics A: Mathematical and Theoretical (2023). doi:10.1088/1751-8121/acc4f5
4. H. Zhan, *The average search probabilities of discrete-time quantum walks*. Quantum Information Processing (2022) 21(9): pp. 336. doi:10.1007/s11128-022-03681-9
5. A. Chan, G. Coutinho, W. Drazen, O. Eisenberg, C. Godsil, M. Kempton, G. Lippner, C. Tamon, H. Zhan, *Fundamentals of fractional revival in graphs*. Linear Algebra and its Applications (2022). doi:10.1016/J.LAA.2022.09.010
6. H. Zhan, *Factoring discrete quantum walks on distance regular graphs into continuous quantum walks*. Linear Algebra and its Applications (2022), 648: pp. 88-103. doi:10.1016/J.LAA.2022.04.017
7. A. Chan, B. Johnson, M. Liu, M. Schmidt, Z. Yin, H. Zhan, *Laplacian pretty good fractional revival*. Discrete Mathematics (2022), 345(10), 112971. doi:10.1016/J.DISC.2022.112971
8. A. Chan, B. Johnson, M. Liu, M. Schmidt, Z. Yin, H. Zhan, *Laplacian fractional revival on graphs*. Electronic Journal of Combinatorics (2021) 28(3): P3.22.
9. A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan, *Fractional revival and association schemes*. Discrete Mathematics (2020) 343(11), 112018. doi:10.1016/j.disc.2020.112018
10. L. Vinet, H. Zhan, *Perfect state transfer on weighted graphs of the Johnson scheme*. Letter in Mathematical Physics (2020). doi.org/10.1007/s11005-020-01298-6
11. H. Zhan, *Quantum walks on embeddings*. Journal of Algebraic Combinatorics (2020). doi.org/10.1007/s10801-020-00958-z
12. H. Zhan, *An infinite family of circulant graphs with perfect state transfer in discrete quantum walks*. Quantum Information Processing (2019) 18(12): pp. 369.

13. G. Coutinho, L. Vinet, H. Zhan, A. Zhedanov. *Perfect state transfer in a spin chain without mirror symmetry*. Journal of Physics A: Mathematical and Theoretical (2019) 52(45), pp. 455302.
14. C. Godsil, H. Zhan, *Discrete-time quantum walks and graph structures*. Journal of Combinatorial Theory, Series A (2019), pp. 181–212.
15. A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan, *Quantum fractional revival on graphs*. Discrete Applied Mathematics (2019) 269, pp. 86-98.
16. G. Coutinho, C. Godsil, K. Guo, H. Zhan, *A new perspective on the average mixing matrix*. Electronic Journal of Combinatorics (2018) 25(4): P4.14.
17. C. Godsil, H. Zhan, *Uniform mixing on Cayley graphs*. Electronic Journal of Combinatorics (2017) 24(3): P3.20.
18. G. Coutinho, C. Godsil, M. Shirazi, H. Zhan, *Equiangular lines and covers of the complete graph*. Linear Algebra and its Applications (2016) 488: pp. 264-283.
19. R. Alvir, S. Dever, B. Lovitz, J. Myer, C. Tamon, Y. Xu, H. Zhan. *Perfect state transfer in Laplacian quantum walk*. Journal of Algebraic Combinatorics (2016) 43(4): pp. 801-826.

Preprints

20. H. Zhan. *Simple quantum coins enable pretty good state transfer on every hypercube*. arXiv:2412.20753 (2024).
21. A. Mohan, H. Zhan. *Discrete quantum walks with marked vertices and their average vertex mixing matrix*. arXiv:2411.16676 (2024).
22. H. Kumar, B. Mohar, S. Pragada, H. Zhan. *On the second largest adjacency eigenvalue of trees with given diameter*. arXiv:2409.01431 (2024).
23. H. Zhan. *ϵ -uniform mixing in discrete quantum walks*. arXiv:2311.18797 (2023).
24. H. Kumar, B. Mohar, S. Pragada, H. Zhan. *Subdivision and graph eigenvalues*. arXiv:2303.10488 (2023).

MENTORING EXPERIENCE

Graduate Research Programs

- WPI Graduate Independent Studies Project Jan - Present, 2024
Supervised a graduate research project on discrete quantum walks.

Undergraduate Research Programs

- WPI STAR Summer Research Training Program May - Aug, 2024
Supervising an undergraduate research project on continuous quantum walks.
- Fields Undergraduate Summer Research Program Jul - Aug, 2020
Co-supervised an undergraduate research project with Ada Chan on Laplacian fractional revival at the Fields Institute. This results in two papers [6] and [7]. One student was awarded the Book Prize by the Fields Institute.
- University of Waterloo May - Aug, 2018
Mentored one undergraduate student under the supervision of Chris Godsil. The student extended my results in the paper [10], and developed the theory of a new quantum walk called the vertex-face walk. This leads to the joint work [19].

PRESENTATIONS **Invited Talks**

1. *Limiting behavior of coined quantum walks with marked vertices.* In: CMS Winter Meeting, Kwantlen Polytechnic University, Vancouver, BC, Canada, November 29 - December 2, 2024.
2. *Recent progress in coined quantum walks.* In: Movement and Symmetry in Graphs, Banff International Research Station for Mathematical Innovation and Discovery, Banff, AB, Canada, November 24 - 29, 2024.
3. *Walking on graphs with a quantum coin.* In: Math Colloquium, University of Manitoba, Winnipeg, MB, Canada, November 21, 2024.
4. *Laplacian fractional revival on graphs.* In: Math Colloquium, University of Hartford, Hartford, CT, United States, October 4, 2024.
5. *Epsilon-uniform mixing on strongly regular graphs via coined quantum walks.* In: 05C50 Online, University of Manitoba, Winnipeg, MB, Canada, September 13, 2024.
6. *Generating quantum uniform mixing in association schemes.* In: CMS Summer Meeting, University of Saskatchewan, Saskatoon, SK, Canada, May 31 - June 3, 2024.
7. *Discrete quantum walks in schemes.* In: AMS Sectional Meeting, University of Wisconsin-Milwaukee, Milwaukee, WI, United States, April 20 - 21, 2024.
8. *Quantum walks: from continuous to discrete.* In: Godsil75, University of Waterloo, Waterloo, ON, Canada, March 15 - 17, 2024.
9. *Strongly cospectral vertices and their phantom mates.* In: Joint Mathematics Meetings, Moscone Center, San Francisco, CA, United States, January 3 - 6, 2024.
10. *Quantum search: an averaging perspective.* In: Joint Mathematics Meetings, Moscone Center, San Francisco, CA, United States, January 3 - 6, 2024.
11. *Spectra of line digraphs and their applications.* In: CMS Winter Meeting, Montréal, QC, Canada, December 1 - 4, 2023.
12. *How graph spectra determine the behavior of discrete quantum walks.* In: 10th International Workshop of Quantum Simulation and Quantum Walks, Tsukuba International Congress Center, Tsukuba, Japan, November 10 - 12, 2023.
13. *On the second largest eigenvalue of a tree.* In: On-line Combinatorics Seminar, University of Wisconsin-Madison, Madison, WI, October 23, 2023.
14. *Some recent results in quantum walks.* In: Algebraic structures and special functions in theoretical physics, Ghent University, Ghent, Belgium, June 26 - 30, 2023.
15. *Discrete quantum walks in association schemes.* In: 10th Slovenian Conference on Graph Theory, Kranjska Gora, Slovenia, June 18 - 24, 2023.
16. *Some open problems in discrete quantum walks.* In: Canadian Discrete and Algorithmic Mathematics Conference, University of Manitoba and University of Winnipeg, Winnipeg, MB, Canada, June 5 - 8, 2023.
17. *Discrete quantum walks on distance regular graphs.* In: Canadian Discrete and Algorithmic Mathematics Conference, University of Manitoba and University of Winnipeg, Winnipeg, MB, Canada, June 5 - 8, 2023.

18. *The second largest eigenvalue of a tree.* In: CMS Winter Meeting, Chelsea Hotel, Toronto, ON, Canada, December 2 - 5, 2022.
19. *The effect of marking vertices in discrete quantum walks.* In: Graph Theory, Algebraic Combinatorics and Mathematical Physics, Centre de Recherches Mathématiques, July 25 - August 19, 2022.
20. *The average search probability in a quantum walk with an oracle.* In: Algebraic Graph Theory Seminar, University of Waterloo, Waterloo, ON, Canada, August 2, 2021.
21. *Arc-reversal quantum walks.* In: Discrete Math Seminar, Simon Fraser University, Vancouver, BC, Canada, February 24, 2021.
22. *DRACKNs and their applications in quantum information.* In: Codes and Expansions, United States, September 8, 2020.
23. *Factoring discrete quantum walks into continuous quantum walks.* In: Algebraic Graph Theory Seminar, University of Waterloo, Waterloo, ON, Canada, August 3, 2020.
24. *Quantum fractional revival.* In: Discrete Math Seminar, University of Delaware, Newark, DE, United States, April 23, 2020.
25. *New advances in quantum walks.* In: AMS Joint Mathematics Meetings, Colorado Convention Center, Denver, CO, United States, January 15 - 18, 2020.
26. *State transfer via orthogonal polynomials.* In: AMS Sectional Meeting, University of Wisconsin-Madison, Madison, WI, United States, September 14 - 15, 2019.
27. *Quantum state transfer in the algebra of the Johnson scheme.* In: CMS Summer Meeting, University of Regina, Regina, SK, Canada, June 7 - 10, 2019.
28. *Some elegant results in algebraic graph theory.* In: Canadian Discrete and Algorithmic Mathematics Conference, Simon Fraser University, Vancouver, BC, Canada, May 28 - 31, 2019.
29. *Quantum walks, orthogonal polynomials, and spectral graph theory.* In: Quantum Walks and Information Tasks, Banff International Research Station for Mathematical Innovation and Discovery, Banff, AB, Canada, April 21 - 26, 2019.
30. *Generating entanglement using quantum walks.* In: David A. Walsh Seminar Series, Clarkson University, Potsdam, NY, United States, February 8, 2019.
31. *Some open problems in discrete quantum walks.* In: Algebraic Graph Theory and Quantum Walks, University of Waterloo, Waterloo, ON, Canada, April 23 - 27, 2018.
32. *Recent progress in discrete quantum walks.* In: AMS Sectional Meeting, Northeastern University, Boston, MA, United States, April 21 - 22, 2018, 2018.
33. *Graph covers and equiangular frames.* In: AMS Sectional Meeting, Ohio State University, Columbus, OH, United States, March 16 - 18, 2018.
34. *From covers to tight frames.* In: AMS Sectional Meeting, College of Charleston, Charleston, SC, United States, March 10 - 12, 2017.
35. *Spectra of discrete quantum walks.* In: CMS Summer Meeting, University of Alberta, Edmonton, AB, Canada, June 24 - 27, 2016.

36. *Lines and covers of complete graphs 2*. In: Systems of Lines: Applications of Algebraic Combinatorics, Worcester Polytechnic Institute, Worcester, MA, United States, August 10 - 14, 2015.
37. *Some open problems in uniform mixing*. In: Summer Research Program, Clarkson University, Potsdam, NY, United States, July 20, 2015.

Mini-Course

38. *Introduction to discrete quantum walks*. In: CMS Winter Meeting, Chelsea Hotel, Toronto, ON, Canada, December 2 - 5, 2022.

Contributed Talks

39. *Average mixing in discrete quantum walks with marked vertices*. In: AMS Fall Meeting, Albany, MA, USA, October 19 - 20, 2024.
40. *ϵ -uniform mixing in discrete quantum walks*. In: CMS Winter Meeting, Montréal, QC, Canada, December 1 - 4, 2023.
41. *Discrete quantum walk search on graphs*. In: Coast Combinatorics Conference 2023, SFU Harbour Centre, Vancouver, BC, Canada, March 4 - 5, 2023.
42. *How far can the quantum walker go*. In: 9th International Conference on Quantum Simulation and Quantum Walks, Centre International de Rencontres Mathématiques, Marseille, Bouches-du-Rhone, France, January 20 - 24, 2020.
43. *Discrete quantum walks on Cayley graphs*. In: CMS Winter Meeting, Chelsea Hotel, Toronto, ON, Canada, December 6 - 9, 2019.
44. *The vertex-face walk*. In: Finite Geometry and Extremal Combinatorics, University of Delaware, Newark, DE, United States, August 21 - 24, 2019.
45. *Combinatorial aspects of quantum walks*. In: Prairie Discrete Math Workshop, Brandon University, Brandon, MB, Canada, June 12 - 15, 2018.
46. *Discrete-time quantum walks and graph embeddings*. In: CMS Winter Meeting, University of Waterloo, Waterloo, ON, Canada, December 8- 11, 2017.
47. *Quantum walks and mixing*. In: Algebraic and Extremal Graph Theory, University of Delaware, Newark, DE, United States, August 7 - 10, 2017.
48. *Discrete-time quantum walks and graph structures*. In: Canadian Discrete and Algorithmic Mathematics Conference, Ryerson University, Toronto, ON, Canada, June 12 - 15, 2017.
49. *Uniform mixing in quantum walks*. In: 22nd Ontario Combinatorics Workshop, York University, Toronto, ON, Canada, May 16 - 17, 2014.

TEACHING EXPERIENCE

Worcester Polytechnic Institute, Worcester, MA, USA

- Instructor Spring 2024
CS 503: Foundations of Computer Science (class size: 25)
- Instructor Spring D-term 2024
CS 4123: Theory of Computation (class size: 40)

Simon Fraser University, Vancouver, BC, Canada

- Instructor Spring 2023
MACM 201: Discrete Mathematics II (class size: 180)
- Instructor Summer 2022
MATH 240: Algebra I: Linear Algebra (class size: 90)

Online Graduate Courses, International

- Instructor Winter 2021
Combinatorics and Quantum Walks (class size: 20; recordings available)

York University, Toronto, ON, Canada

- Instructor Winter 2020, Fall 2020
MATH 1014: Applied Calculus II (class size: 180 - 280)

University of Waterloo, Waterloo, ON, Canada

- Instructor Winter 2018
MATH 135: Algebra for Honors Mathematics (class size: 60)
- Substitute Instructor Winter 2017
CO 444/644: Algebraic Graph Theory (class size: 20)

**FUNDING,
AWARDS AND
DISTINCTIONS**

Federal Research Grants

- NSF CRII: FET: Quantum Advantages through Discrete Quantum Walks 2024 - 2026
Award number: 2348399
Amount: \$174420.00

Funding for Conferences Organized

- Algebraic Graph Theory and Quantum Information 2021

Type and Source	Amount in CAD	Purpose
Fields Institute	\$14250	travel funding for non-US invited speakers, at least \$5000 towards ECRs, females, postdocs, students
NSF, via Fields Institute	\$10000	travel funding for US invited speakers, at least 2/3 towards ECRs, females, postdocs, students
York University, via Office of Vice-President Research and Innovation	\$2000	registration fees for students and postdocs

Table 1: Funding for Algebraic Graph Theory and Quantum Information

Awards and Distinctions from University of Waterloo

- *University Finalist for the Governor General's Gold Medal* 2019
Purpose: a medal awarded to nominees for the Governor General's Gold Medal

- *Inaugural Mathematics Doctoral Prize, First Prize* 2019
Purpose: a prize awarded to recognize the achievement of graduating doctoral students in the Faculty of Mathematics
Amount: \$1500
- *Outstanding Achievement in Graduate Studies* 2015
Purpose: an honor awarded to three University of Waterloo Master's students for their outstanding achievement in graduate studies
- *Cotton Family Women in Mathematics Graduate Scholarship* 2014, 2016, 2017
Purpose: a scholarship awarded to a full-time female graduate student on the basis of academic excellence in their studies and research
Amount: \$9000
- *Robin K. Banks Scholarship* 2011 - 2012
Purpose: a scholarship awarded to a full-time student in the Faculty of Arts who have achieved the highest overall average at the end of Year Three
Amount: \$750
- *Faculty of Arts Upper-Year Scholarship* 2011 - 2012
Purpose: a scholarship awarded to outstanding full-time and part-time students in the Faculty of Arts on the basis of overall average
Amount: \$500
- *Dean's Honours List* 2011 - 2012

SERVICE

Conference, Workshop and Seminar Organizer

- Algebraic Graph Theory, CMS Winter Meeting Nov 29 - Dec 2, 2024
Co-organizers: Sooyeong Kim, Hermie Monderde, Bidy Razafimahatratra, Thomas Jung Spier, Xiaohong Zhang
- Quantum Mathematics for Computation, AMS Fall Sectional Meeting Oct 19 - 20, 2024
Co-organizer: Christino Tamon
- QIT Thinking Seminar, Worcester Polytechnic Institute 2024
Co-organizer: William Martin
- Quantum Information on Graphs, Women in Combinatorics Virtual Conference Jul 15 - 16, 2024
Co-organizer: Xiaohong Zhang
- Algebraic Graph Theory for Walking on Graphs, CMS Winter Meeting 2023
Co-organizers: Sooyeong Kim, Hermie Monderde, Christopher Van Bommel, Xiaohong Zhang
- Algebraic Graph Theory and Quantum Information, Fields Institute 2021
Co-organizers: Ada Chan, Gabriel Coutinho, Krystal Guo, Christino Tamon, Luc Vinet
- Quantum Information on Graphs, CMS Winter Meeting Dec 6 - 9, 2019
Co-organizers: Ada Chan, Christino Tamon
- Discrete Mathematics Seminar, York University Fall 2019
Co-organizer: Justin M. Troyka

Seminar Chair

- Algebraic Graph Theory Seminar, University of Waterloo Spring 2016, Fall 2017

Journal Guest Editor

- Special Issue “Numerical Analysis, Spectral Graph Theory, Orthogonal Polynomials, and Quantum Algorithms” of Philosophical Transactions A 2024
Editors: Anastasiia Minenkova, Gamal Mograby

Journal Reviewer

- Quantum Information Processing 2022 - 2023
- Discrete Mathematics 2021 - 2023
- Linear Algebra and Its Applications 2016, 2019 - 2023
- Linear and Multilinear Algebra 2019 - 2020, 2023
- Algebraic Combinatorics 2022
- Journal of Combinatorial Theory, Series A 2021
- Electronic Journal of Combinatorics 2017 - 2021
- Communications in Algebra 2019
- Journal of Physics A: Mathematical and Theoretical 2019

Conference Reviewer

- Sampling Theory and Applications 2019
- International Colloquium on Automata, Languages and Programming 2018